

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-7	Team No-20	Project Title-Blood Bank and Donor Locator App
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1. ABSTRACT

Blood Bank and Donor Locator App is an Android-based application designed to connect blood donors with recipients who require blood during emergencies or planned medical procedures. Finding a suitable blood donor quickly can be difficult, especially when there is limited information about donor availability and location. The proposed application addresses this problem by providing a centralized platform for donor registration, blood requests, and location-based donor searching.

The application allows donors to register and verify their details, including blood group and availability. Recipients can post blood requests and search for compatible donors using geo-location and map-based services. The application uses Google Maps SDK for location-based search and Firebase Realtime Database for storing and managing donor and blood request information. Firebase Cloud Messaging (FCM) is used to send instant notifications to relevant donors when a blood request is posted. The system also maintains donation history and provides an admin panel for managing donors and blood requests. The main objective is to reduce the time required to find compatible blood donors and improve communication between donors and recipients.

2. Team Details

Roll Number(s)	Name of the Student	Signature
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3. Frameworks / Platforms / Software Used:

Framework / Platform / Software	Purpose
Android Studio	Development environment for building the Android application
Java	Programming language for Android development
XML	Designing Android app layouts and user interfaces
Firebase Authentication	User registration and login
Firebase Realtime Database	Storing donor details, blood requests, and donation history
Firebase Cloud Messaging (FCM)	Sending notifications to suitable donors
Google Maps SDK	Displaying maps and locating donors
Google Maps / Location Services	Obtaining the user's location and finding nearby donors
Git / GitHub	Version control and team collaboration
Firebase Console	Managing Firebase services, database, authentication, and notifications

Guide Name and Signature (remarks if any) Dr. Spandana. S